

Assignment 5

- First of all, please upload your code or a link to your repo to the `/code` folder in Whiteboard resources so that the group working on Project B can find inspiration.
- Then choose one of the following assignments to work on.
 - If you worked on the assignment `a` last week, then choose either `b` or `c`.
 - If you worked on the assignment `b` last week, then continue working on it.
 - If you worked on the assignment `c` last week, then continue working on it.

a) Sepsis prediction

Select one of the following models: `GRU-ODE`, `ODE-RNN`, or `ODE-LSTM`, and apply it to the Sepsis dataset. The objective is the early detection of sepsis using clinical ICU data. Specifically, your model must predict whether a patient will develop sepsis within a 6-hour window prior to clinical onset. Submit your code to Whiteboard by 09.12.2025 and present it in class.

b) Open source contribution

Choose one of `GRU-ODE`, `ODE-RNN`, `ODE-LSTM` and create a Jupyter notebook demonstrating how to implement the chosen model using `Difffrax`, `Equinox`, and `Optax`. Make sure your code has a similar style to the ones in <https://github.com/patrick-kidger/diffrax/tree/main/docs/examples>. You can use the Spirals dataset from Assignment 3. Submit your notebook to Whiteboard by 09.12.2025 and present it in class. Prepare this as if it was a contribution, but don't actually submit it to `Difffrax` just yet.

c) Model implementation

Choose one of NCDE or NSDE models from Table 1 in <https://arxiv.org/abs/2502.09885>.

- If you choose an NCDE model: Apply to the Spirals task from Assignment 3.
- If you choose an NSDE model: Apply it to Covid-19 and Lynx-Hare tasks from Assignment 1.

- If you want to implement another Neural Differential equation based model just drop me an email at u.kaya@fu-berlin.de about it.

Submit your code to Whiteboard by 09.12.2025 and present it in class.